Project - 3: Publishing Amazon SNS Messages Privately

Industry: Healthcare

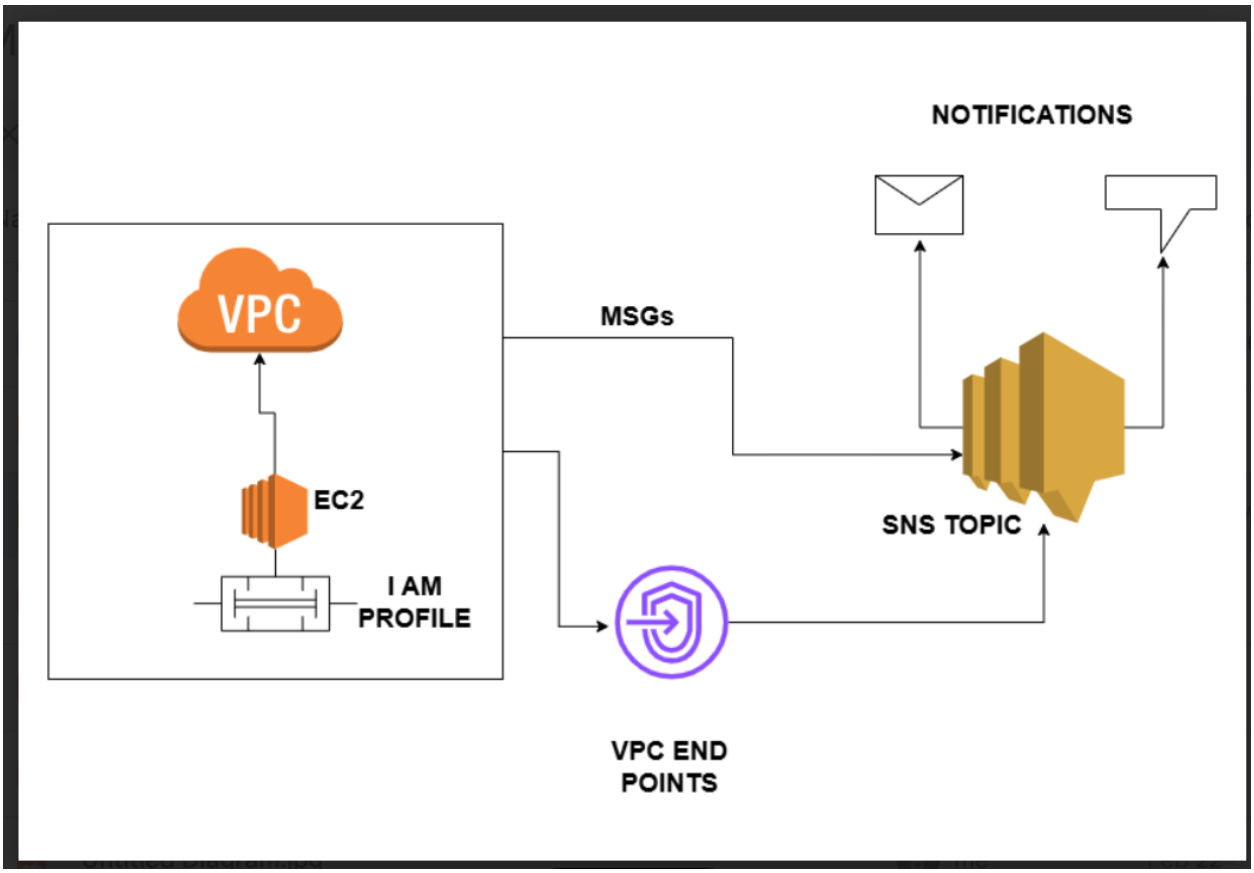
Problem Statement: How to secure patient records online and send it privately to the intended party

Topics: In this project, you will be working on a hospital project to send reports online and develop a platform so the patients can access the reports via mobile and push notifications. You will publish the report to an Amazon SNS keeping it secure and private. Your message will be hosted on an EC2 instance within your Amazon VPC. By publishing the messages privately, you can improve the message delivery and receipt through Amazon SNS.

Highlights: 1. AWS CloudFormation to create a VPC

2. Connect VPC with AWS SNS

3. Publish message privately with SNS



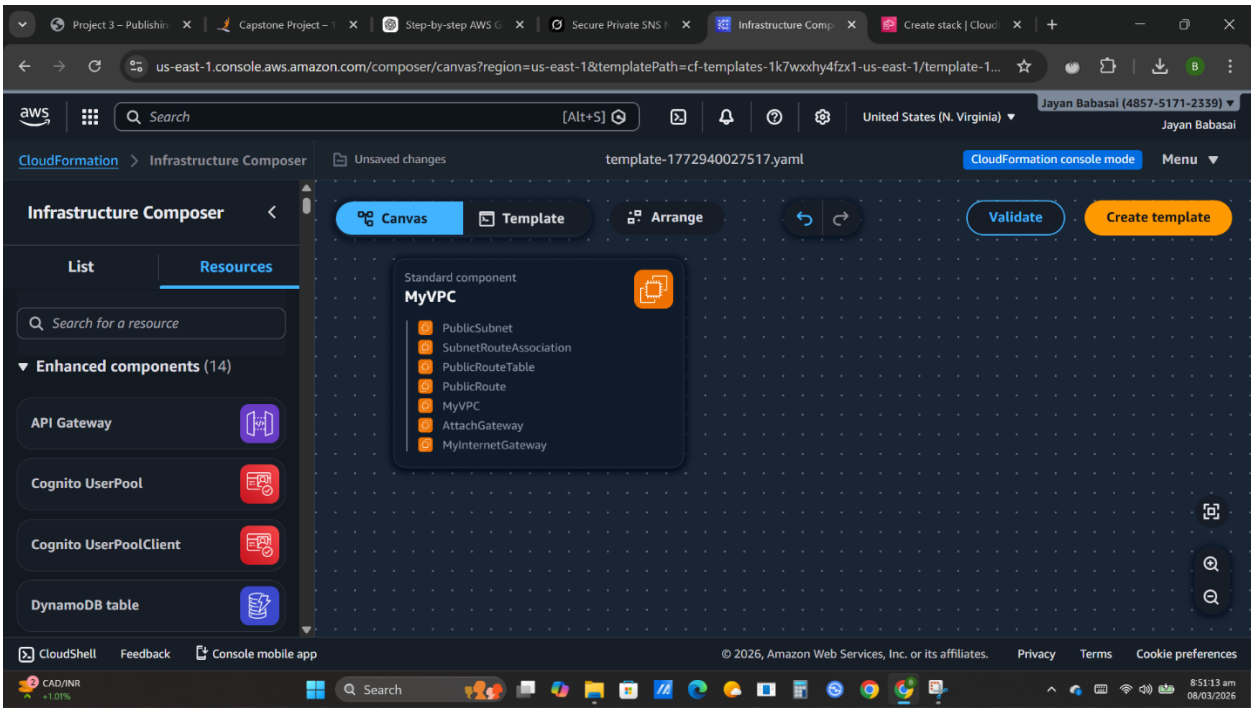
Project Overview:

1. **Objective:**
   Set up a private SNS messaging system inside a VPC using AWS services.

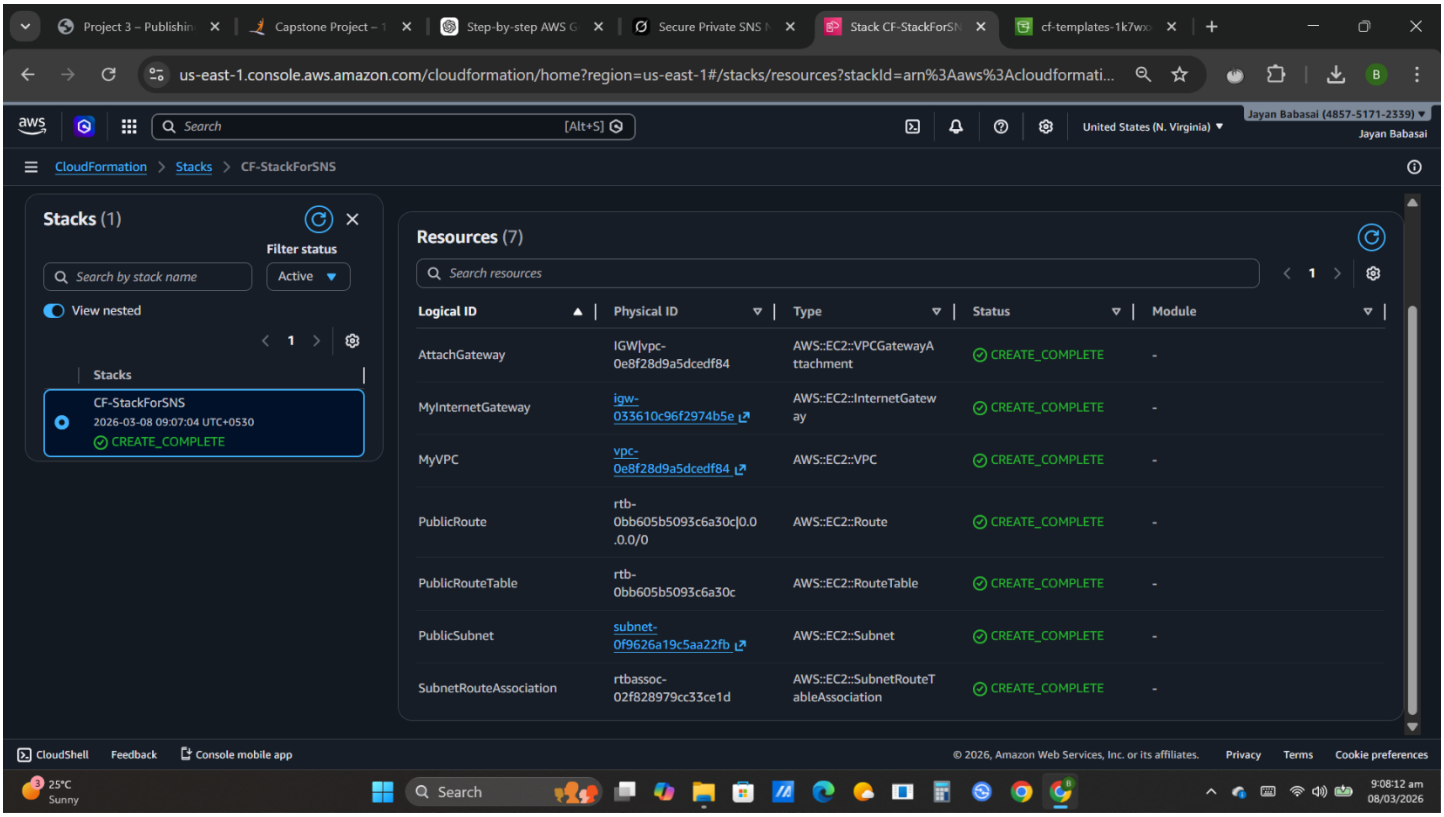
2. **AWS Services Used:**
    - Amazon VPC
    - Amazon EC2
    - Amazon SNS
    - AWS CloudFormation
3. **Steps Completed:**
    - Created a **VPC** with a public subnet using **CloudFormation**.
    - Launched an **EC2 instance** in the VPC.
    - Created an **SNS topic** named hospital-reports.
    - Configured a **VPC Endpoint** for SNS for private access.
    - Installed and configured **AWS CLI** on EC2.
    - Published SNS messages **privately** from the EC2 instance.
4. **Outcome:**
   Messages are successfully published **privately** within the VPC, ensuring secure communication between EC2 and SNS.
5. **Key Learning:**
    - Creating private connections between VPC and AWS services.
    - Automating infrastructure using CloudFormation.
    - Managing AWS resources securely using CLI.

Step-1: Create VPC using CloudFormation

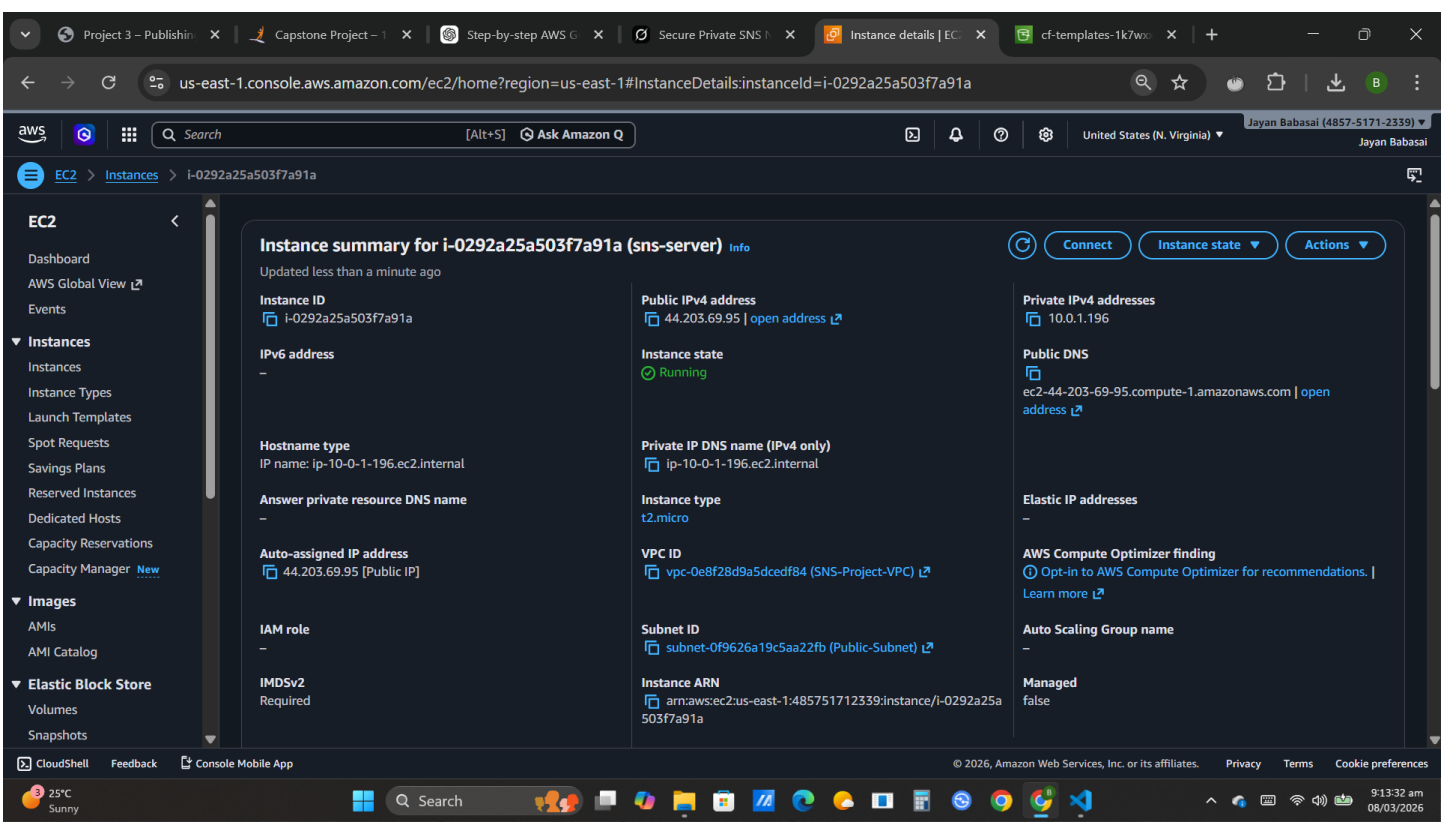
1. Search **CloudFormation**
2. Click **Create Stack**
3. Choose **With new resources (standard)**
4. In **Template Source**




```yaml
AWSTemplateFormatVersion: '2010-09-09'
Description: Simple VPC for SNS Private Project

Resources:

  MyVPC:
    Type: AWS::EC2::VPC
    Properties:
      CidrBlock: 10.0.0.0/16
      EnableDnsSupport: true
      EnableDnsHostnames: true
      Tags:
        - Key: Name
          Value: SNS-Project-VPC

  MyInternetGateway:
    Type: AWS::EC2::InternetGateway
    Properties:
      Tags:
        - Key: Name
          Value: SNS-Project-IGW

  AttachGateway:
    Type: AWS::EC2::VPCGatewayAttachment
    Properties:
      VpcId: !Ref MyVPC
      InternetGatewayId: !Ref MyInternetGateway

  PublicSubnet:
    Type: AWS::EC2::Subnet
    Properties:
      VpcId: !Ref MyVPC
      CidrBlock: 10.0.1.0/24
      MapPublicIpOnLaunch: true
      AvailabilityZone: !Select
        - 0
        - !GetAZs ''
      Tags:
        - Key: Name
          Value: Public-Subnet

  PublicRouteTable:
    Type: AWS::EC2::RouteTable
    Properties:
      VpcId: !Ref MyVPC
      Tags:
        - Key: Name
          Value: Public-RouteTable

  PublicRoute:
    Type: AWS::EC2::Route
    DependsOn: AttachGateway
    Properties:
      RouteTableId: !Ref PublicRouteTable
      DestinationCidrBlock: 0.0.0.0/0
      GatewayId: !Ref MyInternetGateway

  SubnetRouteAssociation:
    Type: AWS::EC2::SubnetRouteTableAssociation
    Properties:
      SubnetId: !Ref PublicSubnet
      RouteTableId: !Ref PublicRouteTable
```



Step 2 — Launch EC2 Instance & Connect & Install AWS CLI & Config aws console



```
  ,     #_
  ~\_  ####_        Amazon Linux 2023
 ~~  \_#####\
 ~~     \###|
 ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
  ~~       V~' '->
   ~~~         /
     ~~._.   _/
        _/ _/
      _/m/'
[ec2-user@ip-10-0-1-196 ~]$ sudo yum update -y
```

i-0292a25a503f7a91a (sns-server)

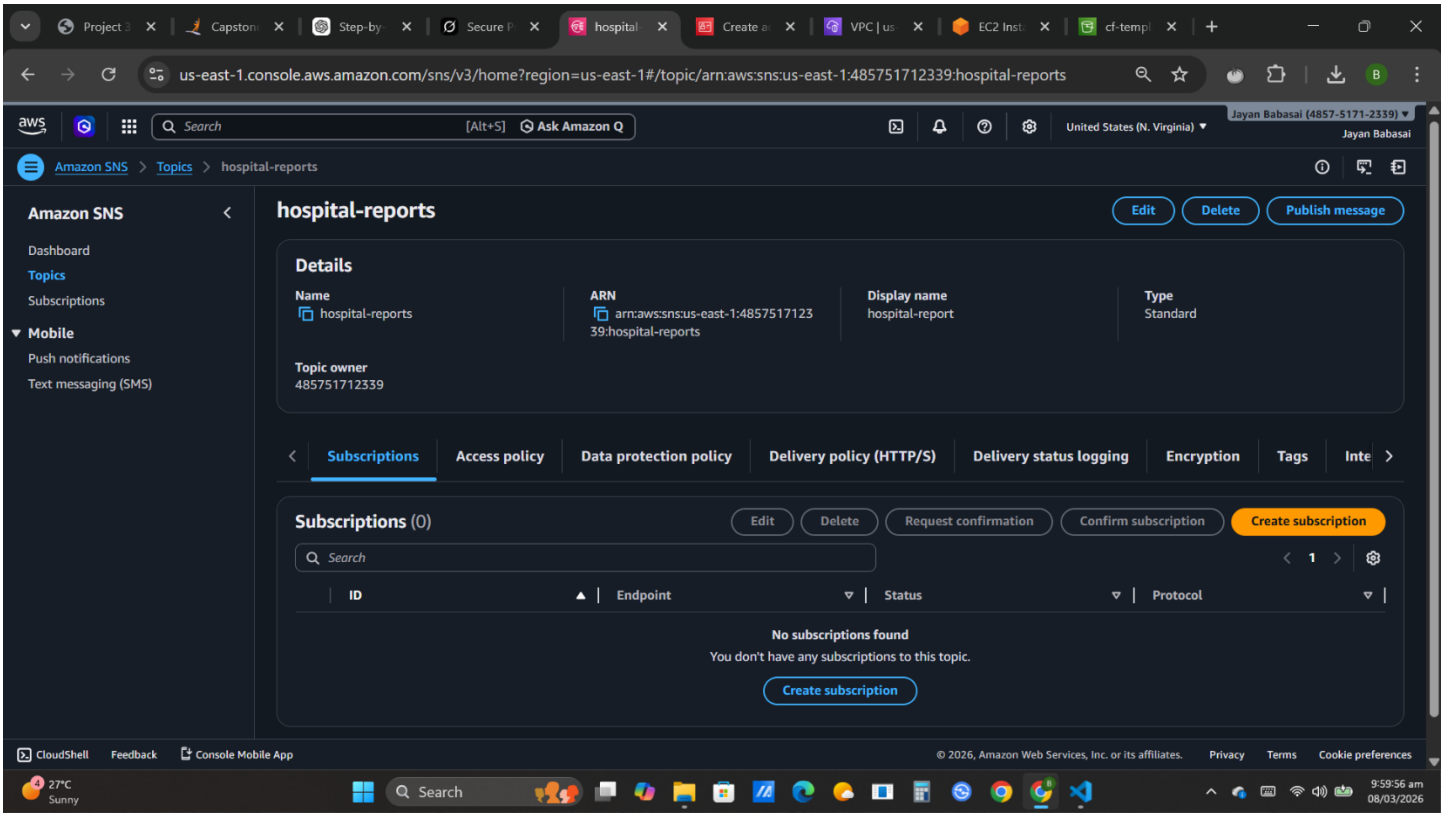
PublicIPs: 44.203.69.95 PrivateIPs: 10.0.1.196

```
  ,     #_
  ~\_  ####_        Amazon Linux 2023
 ~~  \_#####\
 ~~     \###|
 ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
  ~~       V~' '->
   ~~~         /
     ~~._.   _/
        _/ _/
      _/m/'
[ec2-user@ip-10-0-1-196 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository                                    214 kB/s |  30 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-1-196 ~]$ sudo yum install aws-cli -y
Last metadata expiration check: 0:08:26 ago on Sun Mar  8 03:45:28 2026.
Package awscli-2-2.33.15-1.amzn2023.0.1.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-1-196 ~]$ aws --version
aws-cli/2.33.15 Python/3.9.25 Linux/6.1.163-186.299.amzn2023.x86_64 source/x86_64.amzn.2023
[ec2-user@ip-10-0-1-196 ~]$ aws configure
AWS Access Key ID [None]: AKIAXCGIO5JJRWTQZZ5E
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
[ec2-user@ip-10-0-1-196 ~]$
```

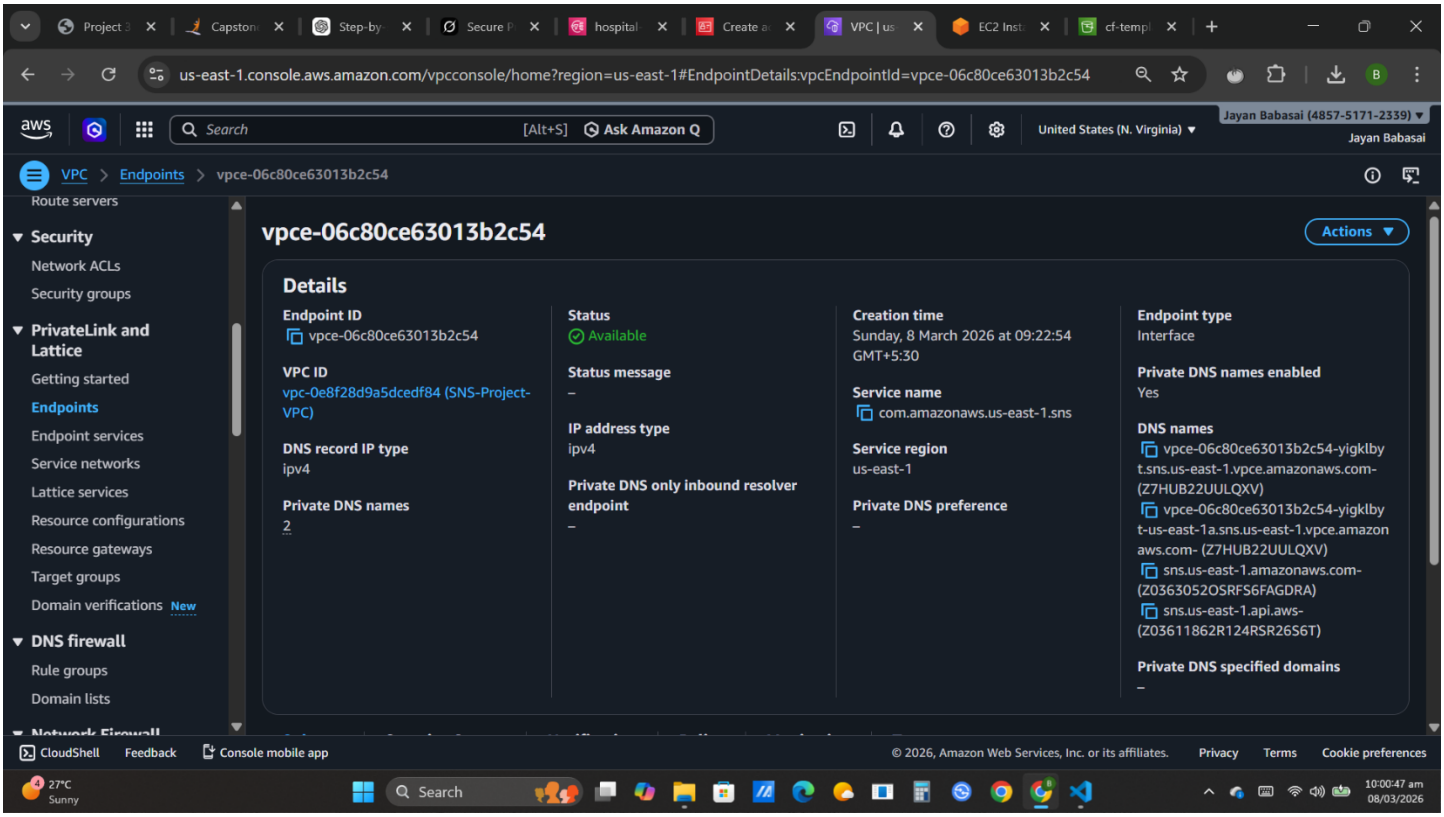
i-0292a25a503f7a91a (sns-server)

PublicIPs: 44.203.69.95 PrivateIPs: 10.0.1.196

Step 3 — Create SNS Topic



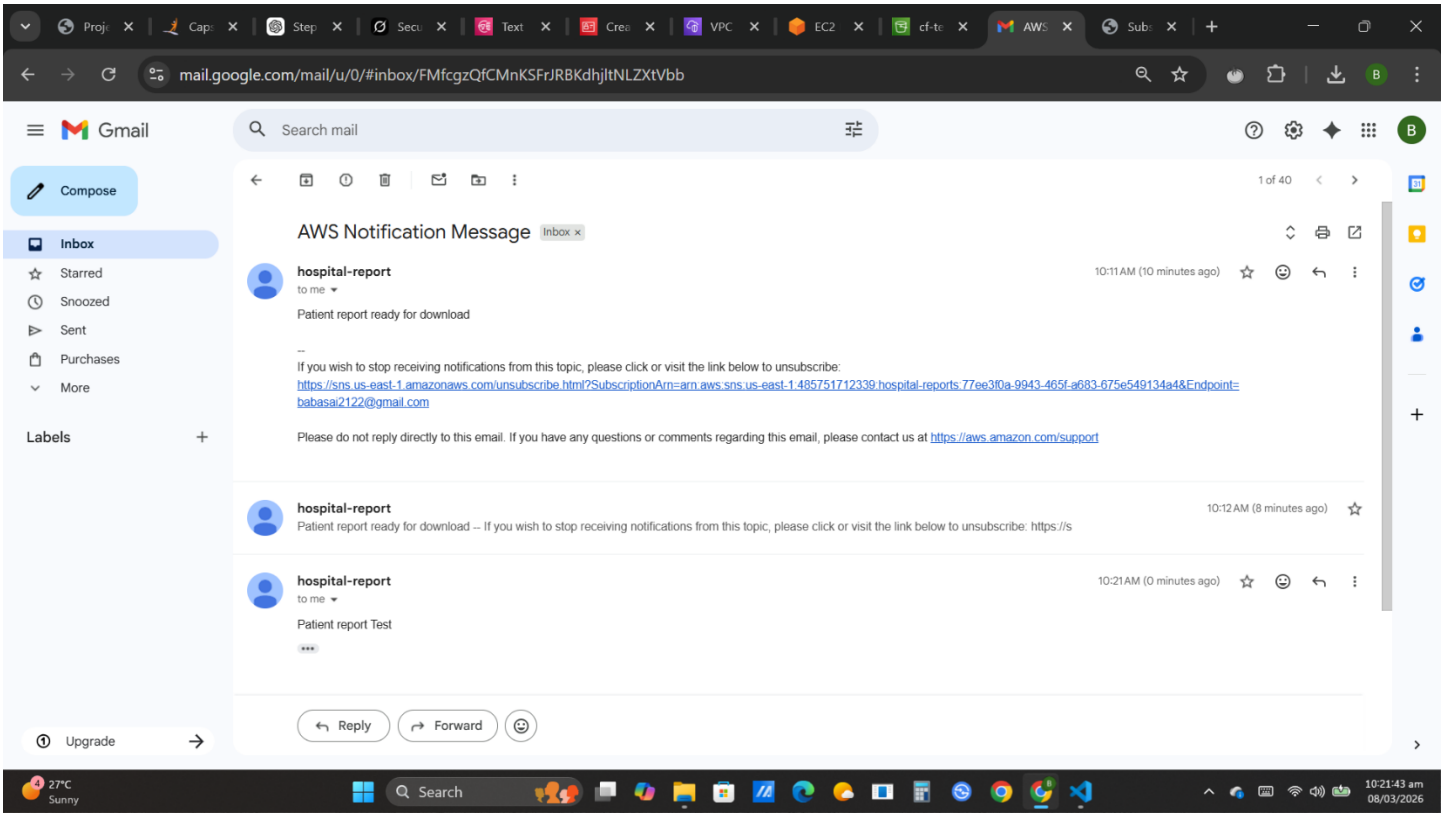
Step 4 — Create VPC Endpoint for SNS (Private Connection)



Step 9 — Publish Private SNS Message

Example:

Run command in EC2:

arn:aws:sns:us-east-1:485751712339:hospital-reports

```
aws sns publish \
--topic-arn arn:aws:sns:us-east-1:485751712339:hospital-reports \
--message "Patient report ready for download"
```